

BIOLOGICAL EVALUATION OF IRON (II) AND ZINC (II) — AMINO ACID COMPLEXES

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ABSTRACT

Essential metalloelements are usually present in the human body as low-molecular-weight protein complexes and are required for various vital physiological and metabolic functions. The bioavailability of these elements depends on their absorption from food sources. Their lower or higher amounts adversely affect the biological functions.

Iron (II) and zinc (II) complexes of twelve different amino acids, were synthesized and evaluated for their *in-vivo* absorption in rabbits. All the metal amino acid-complexes showed more absorption as compared to the respective inorganic salts. The iron (II) and zinc (II) phenylalaninates, leucinates and histidinates were more readily absorbed and showed higher metal concentration in sera as compared to the rest of metal-amino acid complexes. The phenylalaninates showed the highest absorption followed by leucinates and histidinates.

Keywords: Bioavailability, Iron (II), Zinc (II), Amino Acids.

INTRODUCTION

Essential metalloelements are required to regulate certain physiological functions but their bioavailability for human body depends on different aspects, i.e. presence in different food items, antagonism between different metalloelements, binding with certain complexing agents and their *in-vivo* absorbability. The *in-vivo* absorption of metalloelements is very important with respect to their bioavailability in different clinical conditions. Metalloelements in blood are distributed among metalloproteins, metalloenzymes and in low-molecular-weight complexes¹. At least 10-20% of the metalloelements are in the form of low-molecular-weight amino acid-complexes²⁻⁴. The bioavailability of metalloelements such as iron and zinc is essential for certain important biochemical functions⁵. A number of dietary factors appear to affect the level

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of absorption from gastrointestinal tract i.e. excess of one metal ion may cause decrease of intake/distribution of the other⁶. Their *in-vivo* absorption is very important in a more tolerant form, with increased bioavailability for some essential structural, electrochemical and catalytical functions⁵. The excess as well as deficiency of these elements may cause certain diseases⁵. A variety of transition metal complexes of amino acids have been reported with their preparation methods⁷⁻¹⁰. The copper (II) amino acids preparations are well known for their antibacterial and antiulcer activity^{7,11-14}. We picked the most important metalloelements iron and zinc after copper and prepared their complexes with different amino-acids for *in-vivo* evaluation.

As compared to pure inorganic salts iron (II) and zinc (II)-amino acid complexes are devoid of astringent taste and cause less gastric distress, a positive clinical advantage over their inorganic counterparts, ferrous sulphate and zinc sulphate⁸.

EXPERIMENTAL

Material

Methanol, Iron sulphate, Zinc sulphate of E. Merck. L-alanine, L-arginine, L-aspartate, glycine, L-histidine, L-phenylalanine, L-lysine, L-leucine, L-proline, L-threonine, L-tryptophan, and L-tyrosine and sigma chemical were used. The experimentation was carried out in Albino rabbit.

Method

a) Preparation of metal-amino acid complexes

Iron (II) and zinc (II)-amino acid complexes were prepared in solution form according to the usual method⁵, by mixing 2 mmol of amino acid and 1 mmol of metal in 95% alcohol. The mixture was concentrated by evaporation to one fourth of its original volume and chilled to 0°C. Crystals thus obtained were dried under vacuum over phosphorus pentoxide, and preserved for further investigations.

(b) In-vivo evaluation the complexes

The method to study absorption was used as earlier¹⁵. Adult healthy albino rabbits of either sex weighing 1.5-3 Kg were used. Blood samples (5 ml) were drawn from under the ear artery of rabbits, to find out the initial concentration of metals. The complexes prepared were administered orally to the rabbits in pre-determined dose (equivalent to 100 mg Fe (II) and Zn (II), while the control rabbit with an equal amount of metal salt. The experiments was done in triplicate and blood was again drawn from under the ear artery

Mean Iron sera increase after FeSO_4 and its Amino Acid Complexes adminstration

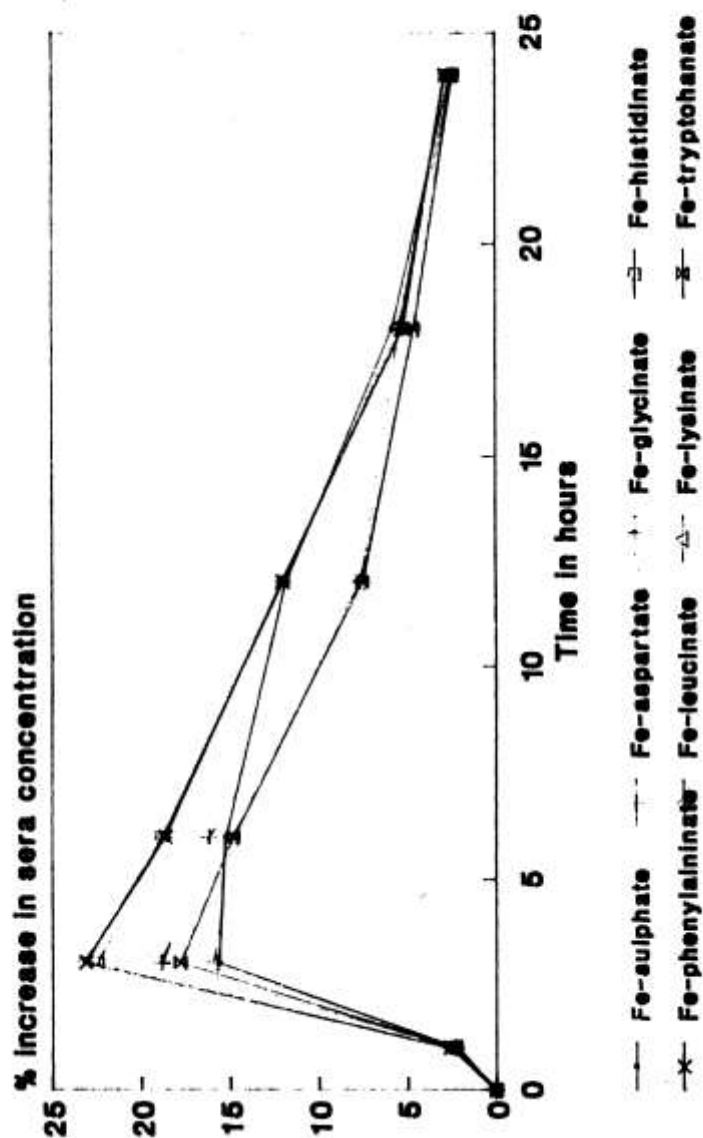


FIGURE No. 1

Mean Zinc sera increase after FeSO_4 and its Amino Acid Complexes administration

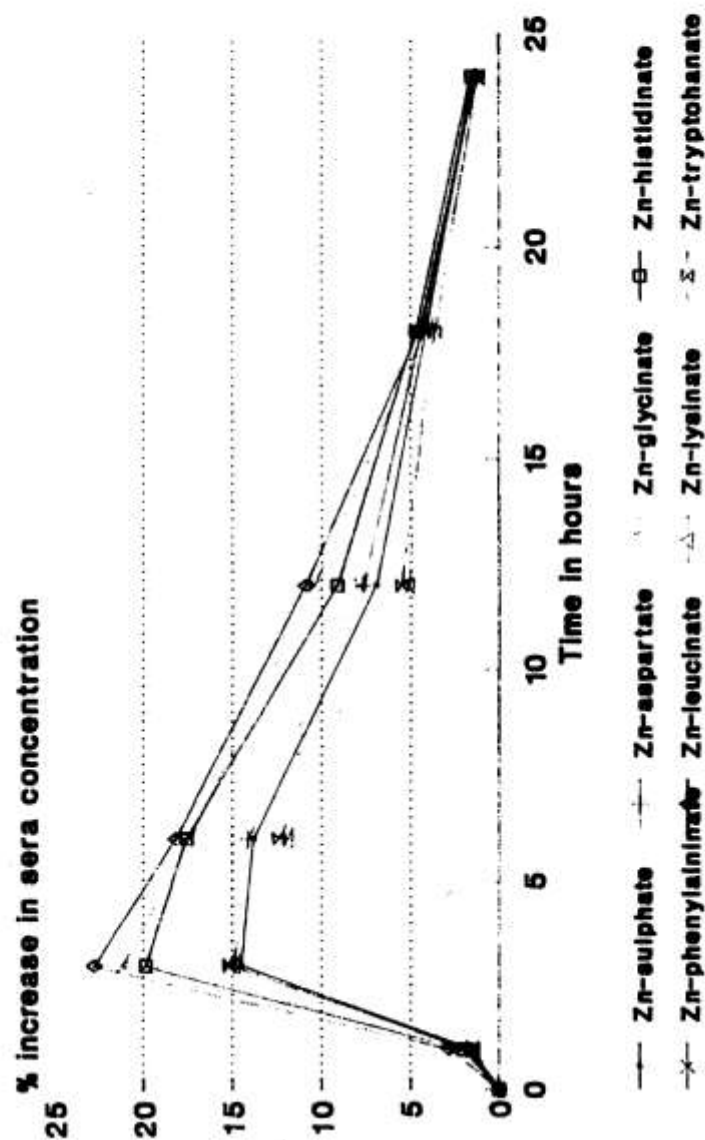


FIGURE No. 2

of each animal at time intervals of 1, 3, 6, 12, 18, 24 hours including controls. Iron and zinc contents were determined by taking the Atomic Absorption spectra.

RESULTS AND DISCUSSION

The sera concentration was plotted against time (Figs. 1 and 2), the values shown in results are the mean of three values. The levels of both metalloelements start increasing in rabbit sera after 1 hour, reach maximum in 3 hours after the oral administration. All metal-amino acid complexes have higher concentrations than the metal salts but concentration found was even more enhanced in the order of phenylalaninates, leucينات and histidinates complexes respectively. The concentration falls gradually with the passage of time and reaches to little higher than its initial concentration after 18 hours and to normal level in 24 hour in each case. The values obtained with alaninates, argininates, prolinates and threoninates are similar to lysinates of both iron and zinc complexes. Therefore these results are not plotted.

The results show that Iron concentration rises relatively faster than the zinc concentration. The concentration fall rate was little fast in case of metal salts and aspartates, glycinates and tryptophanates complexes. The concentration of iron was slightly higher compared to zinc.

The results show that metal-amino acid complexes are more biocompatible and thus can help in increasing the absorability of essential metalloelements. The phenylalaninates, leucينات and histidinates complexes are more rapidly absorbed than other complexes. They can be more beneficial to increase the body content of a metalloelement in higher amount, which help in treating certain diseases i.e. zinc is required for wound healing, burns, surgical incisions, ulcers and growth and development of gonadal organisms¹³, where as iron in injury, bleeding, pregnancy and menstruation cycle¹³.

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